9. Testing Hypotheses and Assessing Goodness of Fit

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SS26 MTH344: Mathematical Statistics

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Lecture Topics

M1. Review of Probability Theory

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Lecture Topics

- In M2, we studied estimation, the process of finding the most plausible probabilistic models given data. In M3, we will study the statistical framework for examining whether the data are consistent with a chosen model.
- See Chapter 9 in **Mathematical Statistics and Data Analysis [MSDA]** (Rice, 2007) for details. Selected examples and text excerpts are reproduced from [MSDA] for educational use only under fair use.
- This course presents content mostly following the textbooks, but also provides substantial unique and advanced material.
- Sections 9.6, 9.7, 9.8, and 9.9 are assigned as reading materials and will not be covered in this course.

Lecture Topics: Statistical Hypothesis Testing

- Consider a scenario where we have two coins:
  1. **Coin 0**: A fair coin with a probability of .5 of getting heads.
  2. **Coin 1**: An unfair coin with a probability of .7 of getting heads.
- Suppose someone selects one of the two coins (with unknown probabilities for each coin) and flips the chosen coin 10 times, resulting in 6 heads.
- **Q**: Based on observing 6 heads, how can we determine whether the chosen coin is Coin 0 or Coin 1?

Lecture Topics: Statistical Hypothesis Testing

- We first briefly review the Bayesian inference framework.
- Consider two competing *hypotheses*:

$$H_0 : \text{Coin 0 (fair) was chosen} \quad \text{vs.} \quad H_1 : \text{Coin 1 (unfair) was chosen} \tag{1}$$

- The Bayesian framework assigns probabilities to each hypothesis prior to observing data (or *evidence*). For instance, $P(H_0)$ and $P(H_1)$ denote the probabilities of selecting each coin, representing out prior degrees of belief in these hypotheses.
- It then updates these probabilities after observing the data to evaluate the posterior odds:[1]

$$\frac{P(H_0 \mid \text{Data})}{P(H_1 \mid \text{Data})}. \tag{2}$$

A higher value indicates stronger evidence in favor of H_0 relative to H_1.

[1] The odds of an event A is defined as $P(A)/P(A^c)$.

Lecture Topics: Statistical Hypothesis Testing

- Let x denote the observed number of heads. Note that Bayes' rule implies:

$$\underbrace{\frac{P(H_0 \mid x)}{P(H_1 \mid x)}}_{\text{Posterior Odds}} = \underbrace{\frac{P(H_0)}{P(H_1)}}_{\text{Prior Odds}} \underbrace{\frac{P(x \mid H_0)}{P(x \mid H_1)}}_{\text{Likelihood Ratio}} . \tag{3}$$

This implies that for any threshold constant c',

$$\underbrace{\frac{P(H_0 \mid x)}{P(H_1 \mid x)}}_{\text{Posterior Odds}} > c' \iff \underbrace{\frac{P(x \mid H_0)}{P(x \mid H_1)}}_{\text{Likelihood Ratio}} > c := c' \left(\underbrace{\frac{P(H_0)}{P(H_1)}}_{\text{Prior Odds}} \right)^{-1} . \tag{4}$$

- Therefore, a higher likelihood ratio implies higher posterior odds. We accept H_0 if the evidence is strong enough to outweigh the prior belief.

Lecture Topics: Statistical Hypothesis Testing

x	0	1	2	3	4	5	6	7	8	9	10
coin 0	.0010	.0098	.0439	.1172	.2051	.2461	.2051	.1172	.0439	.0098	.0010
coin 1	.0000	.0001	.0014	.0090	.0368	.1029	.2001	.2668	.2335	.1211	.0282

x	0	1	2	3	4	5	6	7	8	9	10
$\frac{P(x\|H_0)}{P(x\|H_1)}$	165.4	70.88	30.38	13.02	5.579	2.391	1.025	0.4392	0.1882	0.0807	0.0346

Fig. 9.1 (top) and Fig. 9.2 (bottom) from MSDA

Lecture Topics: Statistical Hypothesis Testing

- Suppose $P(H_0)/P(H_1) = 1$ and we decide to accept H_0 if:

$$\frac{P(H_0 \mid x)}{P(H_1 \mid x)} > 1 \iff \frac{P(x \mid H_0)}{P(x \mid H_1)} > \left(\frac{P(H_0)}{P(H_1)}\right)^{-1} = 1. \tag{5}$$

- The probability of accepting or rejecting H_0 depends on what the true coin was.
- For example, if the chosen coin is Coin 0 (fair):

$$P(\text{accept } H_0 \mid H_0) = P\left(\left\{x : \frac{P(x \mid H_0)}{P(x \mid H_1)} > 1\right\} \mid H_0\right) = P(X < 7 \mid H_0) = .82. \tag{6}$$

Consequently, the probability of making an incorrect decision (rejecting H_0 when it is true) is $P(\text{reject } H_0 \mid H_0) = .18$.

Lecture Topics: Statistical Hypothesis Testing

- In contrast, if the chosen coin is Coin 1 (unfair):

$$P(\text{reject } H_0 \mid H_1) = P\left(\left\{x : \frac{P(x \mid H_0)}{P(x \mid H_1)} \leq 1\right\} \mid H_1\right) = P(X \geq 7 \mid H_1) = .65. \tag{7}$$

Consequently, the probability of making another incorrect decision (accepting H_0 when it is not true) is $P(\text{accept } H_0 \mid H_1) = .35$.

Lecture Topics: Statistical Hypothesis Testing

- Now, suppose $P(H_0)/P(H_1) = 10$ and we decide to accept H_0 if:

$$\frac{P(H_0 \mid x)}{P(H_1 \mid x)} > 1 \iff \frac{P(x \mid H_0)}{P(x \mid H_1)} > \left(\frac{P(H_0)}{P(H_1)}\right)^{-1} = .1. \tag{8}$$

- Then, $P(\text{accept } H_0 \mid H_0) = P(X < 9 \mid H_0) = .99$ and $P(\text{reject } H_0 \mid H_1) = P(X \geq 9 \mid H_1) = .15$.

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The Neyman-Pearson Paradigm

- Unlike the Bayesian approach which assigns degrees of belief to hypotheses, the frequentist framework views the truth of a hypothesis as a fixed state of nature.
- Consequently, H_0 is either strictly true or false (though unknown to us). Probabilities such as $P(\text{accept } H_0 \mid H_0)$ are interpreted as long-run relative frequencies in repeated experiments.
- Even without prior probabilities, we can utilize likelihood ratios to construct decision rules. In the previous example, we can decide to accept H_0 if:

$$\frac{P(x \mid H_0)}{P(x \mid H_1)} > 1 \iff x < 9, \tag{9}$$

  and then compute error probabilities such as $P(\text{reject } H_0 \mid H_1)$, where $X \sim \mathrm{Bin}(10, .5)$ under H_0 and $X \sim \mathrm{Bin}(10, .7)$ under H_1.
- The Neyman-Pearson paradigm establishes the optimality of such likelihood-ratio-based decision rules.

The Neyman-Pearson Paradigm

		Truth (State of Nature)	
		H_0 is True	H_0 is False
Decision	Accept H_0	Correct	Type II Error (β)
	Reject H_0	Type I Error (α)	Correct

- The Neyman-Pearson paradigm treats two competing hypotheses unequally, and refers to H_0 as the *null hypothesis* and H_1 (or H_A) as the *alternative hypothesis*.
- This paradigm aims to find decision rules that maximize $1 - \beta = P(\text{reject } H_0 \mid H_1)$ (*power*) while controlling $\alpha = P(\text{reject } H_0 \mid H_0)$ (*significance level*).

The Neyman-Pearson Paradigm

- **Simple Hypothesis**: A hypothesis is called *simple* if the distribution of the observations is uniquely specified under the hypothesis.
- In the previous example of tossing a coin, both H_0 and H_1 are simple. Under H_0, $X \sim \mathrm{Bin}(10, .5)$, and under H_1, $X \sim \mathrm{Bin}(10, .7)$. This can be formulated as:

$$H_0 : p(x) = p_{\theta_0}(x) \quad \text{vs.} \quad H_1 : p(x) = p_{\theta_1}(x) \tag{10}$$

  where $p_\theta(x)$ represents the PMF of a binomial distribution with parameter vector θ, with $\theta_0 = (10, .5)^T$ and $\theta_1 = (10, .7)^T$.
- **Composite Hypothesis**: A hypothesis is called *composite* if the distribution of the observations is not uniquely specified under the hypothesis.
- For example, consider the following test:[2]

$$H_0 : p(x) = p_{\theta_0}(x) \quad \text{vs.} \quad H_1 : p(x) \neq p_{\theta_0}(x). \tag{11}$$

[2] H_1 allows for any distribution other than the specific PMF defined in H_0.

Likelihood Ratio Test: Coin Example

- Suppose we test two competing simple hypotheses $H_0 : p(x) = p_{\theta_0}(x)$ and $H_1 : p(x) = p_{\theta_1}(x)$, where $\theta_0 = (n, p_0)^T$, $\theta_1 = (n, p_1)^T$, and $p_0 < p_1$.
- The *likelihood ratio test* rejects H_0 for large values of the ratio. For a given threshold $c > 0$, reject H_0 if:

$$\frac{p(x \mid H_1)}{p(x \mid H_0)} = \frac{p_{\theta_1}(x)}{p_{\theta_0}(x)} = \left(\left(\frac{p_1}{1-p_1} \right) / \left(\frac{p_0}{1-p_0} \right) \right)^x \left(\frac{1-p_1}{1-p_0} \right)^n > c. \tag{12}$$

  Since $p_0 < p_1$, this condition is equivalent to $x > c'$ for some constant c'.
- Such decision rules can be formulated as:

$$d(X) = \begin{cases} 1 & \text{if} \quad p_{\theta_1}(X)/p_{\theta_0}(X) > c \\ \gamma & \text{if} \quad p_{\theta_1}(X)/p_{\theta_0}(X) = c \\ 0 & \text{if} \quad p_{\theta_1}(X)/p_{\theta_0}(X) < c. \end{cases} \tag{13}$$

  Here, when $p_{\theta_1}(X)/p_{\theta_0}(X) = c$, we reject H_0 with probability $0 \leq \gamma \leq 1$.

Likelihood Ratio Test: Coin Example

- For example, suppose $\theta_0 = (10, .5)^T$ and $\theta_1 = (10, .7)^T$, and we want to ensure that the significance level is $\mathrm{E}_{\theta_0}[d(X)] = \alpha = .05$.
- Based on our previous derivation, $d(X)$ rejects H_0 for large values of X, implying

$$P_{\theta_0}(X > c') + \gamma P_{\theta_0}(X = c') = \sum_{x=c'+1}^{10} \binom{10}{x} .5^{10} + \gamma \binom{10}{c'} .5^{10} = .05 \tag{14}$$

  for some non-negative integer c'.
- Since $P_{\theta_0}(X \geq 9) \approx .0107$ and $P_{\theta_0}(X \geq 8) \approx .0547$, the desired likelihood ratio-based rule would be:

$$d^{\mathrm{LR}}(X) = \begin{cases} 1 & \text{if} \quad X > 8 \\ \frac{\alpha - P_{\theta_0}(X>8)}{P_{\theta_0}(X=8)} = \frac{.05-.0107}{.0440} \approx .893 & \text{if} \quad X = 8 \\ 0 & \text{if} \quad X < 8. \end{cases} \tag{15}$$

Likelihood Ratio Test: Coin Example

- Under H_1, $X \sim \mathrm{Bin}(10, .7)$ and the power of $d^{\mathrm{LR}}(X)$ is:

$$\begin{aligned} \mathrm{E}_{\theta_1}[d^{\mathrm{LR}}(X)] &= P_{\theta_1}(X > 8) + .893 P_{\theta_1}(X = 8) \\ &= \left(\binom{10}{9} .7^9 .3^1 + \binom{10}{10} .7^{10} \right) + .893 \left(\binom{10}{8} .7^8 .3^2 \right) \\ &\approx .358. \end{aligned} \tag{16}$$

- Given a significance level α, there can be multiple decision rules:
  - Naive Rule: Ignore the data X and simply reject H_0 with a flat probability of .05.[3]

$$\mathrm{E}_{\theta_1}[d^{\mathrm{Naive}}(X)] = .050. \tag{17}$$

  - Reversed Likelihood-Ratio Test: Reject H_0 for *small* values of X ($X < 2$, and with probability .893 if $X = 2$).

$$\mathrm{E}_{\theta_1}[d^{\text{Reversed-LR}}(X)] = P_{\theta_1}(X < 2) + .893 P_{\theta_1}(X = 2) \approx .001. \tag{18}$$

[3] Indeed, even a naive test that completely ignores the data can reject the null hypothesis. Therefore, rejecting H_0 alone may not be fully informative without considering the test's power.

Likelihood Ratio Test: Alpha Particles Example

n	Observed
0–2	18
3	28
4	56
5	105
6	126
7	146
8	164
9	161
10	123
11	101
12	74
13	53
14	23
15	15
16	9
17+	5
	1207

A table in Section 8.2. in [MSDA].

- Recall the $n = 1207$ intervals from the alpha particle experiment in Section 8.2. Suppose the emission counts follow a Poisson distribution with an unknown parameter λ.
- We test a historical emission rate

$$H_0 : \lambda = \lambda_0 = 8 \tag{19}$$

against the alternative hypothesis[a]

$$H_1 : \lambda = \lambda_1 = 8.5. \tag{20}$$

- **Q**: The MLE is $\hat{\lambda} = \bar{x} = 8.392$. Is it large enough to reject H_0 while ensuring an appropriate level of power?

[a] We denote $H_j : \lambda = \lambda_j$ assuming the parameter is *identifiable*, i.e., $f(x; \lambda) = f(x; \lambda')$ for all $x \implies \lambda = \lambda'$.

Likelihood Ratio Test: Alpha Particles Example

- The likelihood ratio for n independent observations $\boldsymbol{x} = (x_1, \ldots, x_n)^T$ is:

$$\frac{p_{\lambda_1}(\boldsymbol{x})}{p_{\lambda_0}(\boldsymbol{x})} = \frac{\prod_{i=1}^n e^{-\lambda_1}\lambda_1^{x_i}/x_i!}{\prod_{i=1}^n e^{-\lambda_0}\lambda_0^{x_i}/x_i!} = e^{-n(\lambda_1-\lambda_0)}\left(\frac{\lambda_1}{\lambda_0}\right)^{\sum_{i=1}^n x_i}. \tag{21}$$

- Since $\lambda_0 < \lambda_1$, the likelihood ratio test is given by:

$$d^{\mathrm{LR}}(\boldsymbol{X}) = \begin{cases} 1 & \text{if} \quad \sum_{i=1}^n X_i > c, \\ \gamma & \text{if} \quad \sum_{i=1}^n X_i = c, \\ 0 & \text{if} \quad \sum_{i=1}^n X_i < c, \end{cases} \tag{22}$$

  for some constant $0 \leq \gamma \leq 1$ and a non-negative integer c.
- Here, $\sum_{i=1}^n X_i$ is referred to as the *test statistic*, c as the *critical value*, and $\{\boldsymbol{x} : \sum_{i=1}^n x_i > c\}$ as the *rejection region*.
- Note that $\sum_{i=1}^n X_i \sim \mathrm{Poi}(n\lambda_0)$ under H_0. Given n, λ_0, and α, we can find γ and c using its PMF.

Likelihood Ratio Test: Alpha Particles Example

- Here, assuming n is sufficiently large, we apply the CLT: Under $H_j : \lambda = \lambda_j$,

$$\frac{\sqrt{n}(\bar{X} - \lambda_j)}{\sqrt{\lambda_j}} \xrightarrow{d} Z, \quad \text{where } Z \sim \mathcal{N}(0,1). \tag{23}$$

- Based on the CLT, the Type I error can be expressed as:

$$\begin{aligned} \mathrm{E}_{\lambda_0}[d^{\mathrm{LR}}(\boldsymbol{X})] = P_{\lambda_0}\left(\sum_{i=1}^{n} X_i > c\right) &= P_{\lambda_0}\left(\frac{\sqrt{n}\left(\bar{X} - \lambda_0\right)}{\sqrt{\lambda_0}} > \frac{\sqrt{n}}{\sqrt{\lambda_0}}\left(\frac{c}{n} - \lambda_0\right)\right) \\ &\approx 1 - \Phi\left(\frac{\sqrt{n}}{\sqrt{\lambda_0}}\left(\frac{c}{n} - \lambda_0\right)\right). \end{aligned} \tag{24}$$

- Given a significance level α, we obtain $c \approx n\lambda_0 + \sqrt{n\lambda_0} z_\alpha$.

Likelihood Ratio Test: Alpha Particles Example

- The corresponding power is:

$$\mathrm{E}_{\lambda_1}[d^{\mathrm{LR}}(\boldsymbol{X})] = P_{\lambda_1}\left(\sum_{i=1}^{n} X_i > c\right) = P_{\lambda_1}\left(\frac{\sqrt{n}\left(\bar{X}-\lambda_1\right)}{\sqrt{\lambda_1}} > \frac{\sqrt{n}}{\sqrt{\lambda_1}}\left(\frac{c}{n}-\lambda_1\right)\right)$$
$$\approx 1-\Phi\left(-\frac{\lambda_1-\lambda_0}{\sqrt{\lambda_1}}\sqrt{n}+\frac{\sqrt{\lambda_0}}{\sqrt{\lambda_1}}z_\alpha\right). \tag{25}$$

- Therefore, to guarantee a target power of at least $1-\beta$ (where β is the Type II error rate), the sample size n must satisfy:[4]

$$n \gtrsim \frac{\lambda_1}{(\lambda_1-\lambda_0)^2}\left(\frac{\sqrt{\lambda_0}}{\sqrt{\lambda_1}}z_\alpha + z_\beta\right)^2. \tag{26}$$

[4] Such experimental design is typically required to obtain approval before conducting experiments.

Likelihood Ratio Test: Alpha Particles Example

- Let $\alpha = .05$ ($z_{.05} \approx 1.645$) and Type II error $\beta = .10$ ($z_{.10} \approx 1.282$). Since $\lambda_0 = 8$ and $\lambda_1 = 8.5$, the approximated lower bound in Equation (26) becomes:

$$\frac{8.5}{(8.5-8)^2}\left(\sqrt{\frac{8}{8.5}}(1.645) + 1.282\right)^2 \approx 281.6. \tag{27}$$

  Since our actual sample size is $n = 1207 \geq 282$, the likelihood ratio test with significance level $\alpha = .05$ is more than enough to ensure the desired power.
- Let's check the test result. Since $\bar{x} = 8.392$,

$$\sum_{i=1}^{n} x_i = n\bar{x} \approx 10129 > c \approx n\lambda_0 + \sqrt{n\lambda_0}z_\alpha = 9656 + \sqrt{9656}(1.645) \approx 9818. \tag{28}$$

  Therefore, we reject H_0 at the significance level $\alpha = .05$.

Likelihood Ratio Test: Alpha Particles Example

- Now, keep $\alpha = .05$ and $\beta = .10$, but suppose we want to consider closer hypotheses:

$$H_0 : \lambda = \lambda_0 = 8 \quad \text{vs.} \quad H_1 : \lambda = \lambda_1 = 8.1. \tag{29}$$

- The approximated lower bound in Equation (26) becomes:

$$\frac{8.1}{(8.1-8)^2}\left(\sqrt{\frac{8}{8.1}}(1.645) + 1.282\right)^2 \approx 6891.2. \tag{30}$$

  Since our actual sample size is $n = 1207 < 6892$, the likelihood ratio test at significance level $\alpha = .05$ is insufficient to ensure the desired power.
- The current power is approximately:

$$1 - \Phi\left(-\frac{8.1-8}{\sqrt{8.1}}\sqrt{1207} + \frac{\sqrt{8}}{\sqrt{8.1}}(1.645)\right) \approx 1 - \Phi(.414) \approx .339, \tag{31}$$

  which is much smaller than $1 - \beta = .90$. The current Type II error is about 66.1%.

Likelihood Ratio Test: Alpha Particles Example

- Nevertheless, the likelihood ratio test rejects H_0, just as in the previous case. Note that the critical value $c \approx n\lambda_0 + \sqrt{n\lambda_0} z_\alpha \approx 9818$ depends only on n, λ_0, and α.
- This rejection makes sense, considering that our observed mean $\bar{x} = 8.392$ is very extreme, far exceeding $\lambda_1 = 8.1$.
- However, what if we observe $\bar{x} = \lambda_1 = 8.1$ and the test fails to reject H_0 ($n\bar{x} \approx 9777 < 9818$)?
- Though such an observation is highly probable when H_1 is true, the conclusion of the likelihood ratio test would be misleading due to its low power (or high Type II error rate).

The Neyman-Pearson Lemma

- A naturally arising question is whether there exists a more powerful test than the likelihood ratio test at a fixed level of significance α.
- <u>Most Powerful Test</u>: Given a random sample $\boldsymbol{X} = (X_1, \ldots, X_n)^T$ from a distribution with PDF $f(\boldsymbol{x})$ (or PMF $p(\boldsymbol{x})$ if discrete), suppose we test two simple hypotheses:

$$H_0 : f(\boldsymbol{x}) = f_0(\boldsymbol{x}) \quad \text{vs.} \quad H_1 : f(\boldsymbol{x}) = f_1(\boldsymbol{x}). \tag{32}$$

A test $d^{\mathrm{MP}}(\boldsymbol{X}) \in [0, 1]$ is defined as the *most powerful test* of level $0 < \alpha < 1$ if it satisfies the following conditions (where E_i denotes the expectation under H_i):

1. $\mathrm{E}_0[d^{\mathrm{MP}}(\boldsymbol{X})] \leq \alpha$,
2. $\mathrm{E}_1[d^{\mathrm{MP}}(\boldsymbol{X})] \geq \mathrm{E}_1[d(\boldsymbol{X})]$ for any test $d(\boldsymbol{X}) \in [0, 1]$ such that $\mathrm{E}_0[d(\boldsymbol{X})] \leq \alpha$.

The Neyman-Pearson Lemma

- **The Neyman-Pearson Lemma**: Suppose we test two simple hypotheses:

$$H_0 : f(\boldsymbol{x}) = f_0(\boldsymbol{x}) \quad \text{vs.} \quad H_1 : f(\boldsymbol{x}) = f_1(\boldsymbol{x}). \tag{33}$$

For a given significance level α, if a *likelihood ratio test* $d^{\mathrm{LR}}(\boldsymbol{X})$ defined as

$$d^{\mathrm{LR}}(\boldsymbol{X}) = \begin{cases} 1 & \text{if} \quad f_1(\boldsymbol{X})/f_0(\boldsymbol{X}) > c \quad \text{or} \quad f_0(\boldsymbol{X}) = 0 \\ \gamma & \text{if} \quad f_1(\boldsymbol{X})/f_0(\boldsymbol{X}) = c \\ 0 & \text{if} \quad f_1(\boldsymbol{X})/f_0(\boldsymbol{X}) < c \end{cases} \tag{34}$$

satisfies $\mathrm{E}_0[d^{\mathrm{LR}}(\boldsymbol{X})] = \alpha$ for some constants $c \geq 0$ and $0 \leq \gamma \leq 1$, then it is the most powerful test of level α.

The Neyman-Pearson Lemma

Exercise 1

Given a random sample $\mathbf{X} = (X_1, \ldots, X_n)^T$ from a distribution with PDF $f(\boldsymbol{x})$ (or PMF $p(\boldsymbol{x})$ if discrete), suppose we test two simple hypotheses:

$$H_0 : f(\boldsymbol{x}) = f_0(\boldsymbol{x}) \quad \text{vs.} \quad H_1 : f(\boldsymbol{x}) = f_1(\boldsymbol{x}). \tag{35}$$

a) For any decision rule $d(\boldsymbol{X})$ such that $0 \leq d(\boldsymbol{X}) \leq 1$ and any constant $c \geq 0$, show that its power can be expressed as:

$$\begin{aligned} \mathrm{E}_1[d(\boldsymbol{X})] = \mathrm{E}_0 &\left[d(\boldsymbol{X}) \left(\frac{f_1(\boldsymbol{X})}{f_0(\boldsymbol{X})} - c \right) I(f_0(\boldsymbol{X}) \neq 0) \right] + c\, \mathrm{E}_0[d(\boldsymbol{X})] \\ &+ \mathrm{E}_1[d(\boldsymbol{X}) I(f_0(\boldsymbol{X}) = 0)]. \end{aligned} \tag{36}$$

The Neyman-Pearson Lemma

The Neyman-Pearson Lemma

b) Let $0 < \alpha < 1$, $c \geq 0$, and $0 \leq \gamma \leq 1$. Show that for any decision rule $d(\boldsymbol{X})$ such that $0 \leq d(\boldsymbol{X}) \leq 1$ and $\mathrm{E}_0[d(\boldsymbol{X})] \leq \alpha$, the following inequality holds:

$$\mathrm{E}_1[d(\boldsymbol{X})] \leq \mathrm{E}_1[d^{\mathrm{LR}}_{c,\gamma}(\boldsymbol{X})] + c\left(\alpha - \mathrm{E}_0[d^{\mathrm{LR}}_{c,\gamma}(\boldsymbol{X})]\right) \tag{37}$$

where $d^{\mathrm{LR}}_{c,\gamma}(\boldsymbol{X})$ represents the *likelihood ratio test* defined as:

$$d^{\mathrm{LR}}_{c,\gamma}(\boldsymbol{X}) = \begin{cases} 1 & \text{if} \quad f_1(\boldsymbol{X})/f_0(\boldsymbol{X}) > c \quad \text{or} \quad f_0(\boldsymbol{X}) = 0 \\ \gamma & \text{if} \quad f_1(\boldsymbol{X})/f_0(\boldsymbol{X}) = c \\ 0 & \text{if} \quad f_1(\boldsymbol{X})/f_0(\boldsymbol{X}) < c \end{cases} \tag{38}$$

The Neyman-Pearson Lemma

The Neyman-Pearson Lemma

- For any significance level $\alpha \in (0, 1)$, there always exist constants c and γ such that the likelihood ratio test d^{LR} has exactly level α.
- Furthermore, every most powerful test must be a likelihood ratio test (except perhaps on a set of points with zero probability under both hypotheses).
- In short, no test is more powerful than the likelihood ratio test when testing simple hypotheses.
- In the previous coin-tossing and alpha particle examples, since both H_0 and H_1 were simple, the derived likelihood ratio tests were indeed the most powerful.

Most Powerful Test: Gaussian

- **Example 9.2.A**: Let $X_1, \ldots, X_n$ be i.i.d. observations from a Gaussian distribution with an unknown μ and a known σ^2. Suppose we test two competing simple hypotheses:

$$H_0 : \theta = \theta_0 = (\mu_0, \sigma^2)^T \quad \text{vs.} \quad H_1 : \theta = \theta_1 = (\mu_1, \sigma^2)^T. \tag{39}$$

By the Neyman-Pearson Lemma, when $\mu_0 < \mu_1$, the likelihood ratio test,

$$d^{\mathrm{LR}}(\boldsymbol{X}) = \begin{cases} 1 & \text{if } \bar{X} > \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}} \\ 0 & \text{otherwise} \end{cases} \tag{40}$$

is the most powerful test of level α. Conversely, when $\mu_0 > \mu_1$, the rejection region is given by $\bar{X} < \mu_0 - z_\alpha \frac{\sigma}{\sqrt{n}}$.

Most Powerful Test: Gaussian

Exercise 2

Let $X_1, \ldots, X_n$ be i.i.d. observations from $N(\mu, \sigma^2)$ with known μ and unknown σ^2. We test simple hypotheses $H_0 : \sigma^2 = \sigma_0^2$ vs. $H_1 : \sigma^2 = \sigma_1^2$ where $\sigma_0^2 < \sigma_1^2$.

a) Show that the most powerful test of level α is given by:

$$d^{\mathrm{MP}}(\boldsymbol{X}) = I\left(\sum_{i=1}^{n}(X_i - \mu)^2 > \sigma_0^2 \chi_\alpha^2(n)\right). \tag{41}$$

Most Powerful Test: Gaussian

Most Powerful Test: Gaussian

b) Let $V \sim \chi^2(n)$. Show that the power of d^{MP} is $1 - F_V\left(\frac{\sigma_0^2}{\sigma_1^2}\chi^2_\alpha(n)\right)$.

Most Powerful Test: Gaussian

c) Show that the minimum sample size n to guarantee a target power of at least $1 - \beta$ is the smallest integer satisfying:

$$\frac{\chi^2_\alpha(n)}{\chi^2_{1-\beta}(n)} \leq \frac{\sigma_1^2}{\sigma_0^2}. \tag{42}$$

Most Powerful Test: Exponential Family

- Consider a single-parameter exponential family with PDF (or PMF):

$$f(x;\eta) = h(x)\exp\left(\eta T(x) - A(\eta)\right) \tag{43}$$

  where $\eta = \eta(\theta)$ is the natural parameter, $T(x)$ is a sufficient statistic, and $A(\eta)$ is the log-partition function.
- Suppose we test $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$ with $\theta_0 < \theta_1$, and $\eta(\theta)$ is a strictly increasing function of θ.
- By the Neyman-Pearson Lemma, the most powerful test of level α is:

$$d^{\mathrm{MP}}(\boldsymbol{X}) = \begin{cases} 1 & \text{if } \sum_{i=1}^n T(X_i) > c \\ \gamma & \text{if } \sum_{i=1}^n T(X_i) = c \\ 0 & \text{if } \sum_{i=1}^n T(X_i) < c \end{cases} \tag{44}$$

  where c and γ are determined by the significance level α.
- The most powerful test depends on the data only through the sufficient statistic $\sum_{i=1}^n T(X_i)$.

Most Powerful Test: Exponential

Exercise 3

Let $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Exp}(\lambda)$ with an unknown λ. We test simple hypotheses $H_0 : \lambda = \lambda_0$ vs. $H_1 : \lambda = \lambda_1$ where $\lambda_0 < \lambda_1$.

a) Show that the most powerful test of level α is given by:

$$d^{\mathrm{MP}}(\boldsymbol{X}) = I\left(\sum_{i=1}^{n} X_i < \frac{1}{2\lambda_0}\chi^2_{1-\alpha}(2n)\right).$$

Most Powerful Test: Exponential

Most Powerful Test: Exponential

b) Let $V \sim \chi^2(2n)$. Show that the power of d^{MP} is $F_V\left(\frac{\lambda_1}{\lambda_0}\chi^2_{1-\alpha}(2n)\right)$.

Most Powerful Test: Exponential

c) Show that the minimum sample size n to guarantee a target power of at least $1 - \beta$ is the smallest integer satisfying:

$$\frac{\chi^2_{1-\alpha}(2n)}{\chi^2_{\beta}(2n)} \geq \frac{\lambda_0}{\lambda_1}.$$

Uniformly Most Powerful Test

- Until now, we have focused on testing simple hypotheses. However, the logic of the Neyman-Pearson Lemma can often be extended to accommodate composite hypotheses.
- **Uniformly Most Powerful Test**: Given a random sample $\mathbf{X} = (X_1, \ldots, X_n)^T$ from a parametric distribution with PDF $f_\theta(\boldsymbol{x})$ (or PMF $p_\theta(\boldsymbol{x})$ if discrete), suppose we test two hypotheses:

$$H_0 : \theta \in \Omega_0 \quad \text{vs.} \quad H_1 : \theta \in \Omega_1, \tag{45}$$

where $\Omega_0 \cap \Omega_1 = \emptyset$. A test $d^{\text{UMP}}(\boldsymbol{X}) \in [0, 1]$ is defined as the *uniformly most powerful (UMP) test* of level $0 < \alpha < 1$ if it satisfies the following conditions:

1. $\sup_{\theta \in \Omega_0} \mathrm{E}_\theta[d^{\text{UMP}}(\boldsymbol{X})] \leq \alpha$
2. $\mathrm{E}_{\theta_1}[d^{\text{UMP}}(\boldsymbol{X})] \geq \mathrm{E}_{\theta_1}[d(\boldsymbol{X})]$ for any $\theta_1 \in \Omega_1$ and any $d(\boldsymbol{X}) \in [0, 1]$ such that $\sup_{\theta \in \Omega_0} \mathrm{E}_\theta[d(\boldsymbol{X})] \leq \alpha$.

Uniformly Most Powerful Test: Gaussian

- Here we consider a *one-sided test*, which is a typical example of composite hypothesis testing. Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with an unknown μ and a known σ^2:

$$H_0 : \mu \leq \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0. \tag{46}$$

- In Example 9.2.A, we showed that, when $\mu_0 < \mu_1$, the most powerful test of level α for $H_0 : \mu = \mu_0$ vs. $H_1 : \mu = \mu_1$ is given by:

$$d^{\text{LR}}(\boldsymbol{X}) = I\left(\bar{X} > \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}\right). \tag{47}$$

- Since the rejection region does not depend on the specific value of μ_1 (as long as $\mu_1 > \mu_0$), d^{LR} is the UMP test of level α for

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0. \tag{48}$$

Uniformly Most Powerful Test: Gaussian

- To extend this to the composite null hypothesis, first note that the power function:

$$\begin{aligned} \mathrm{E}_\mu[d^{\mathrm{LR}}(\boldsymbol{X})] = P_\mu\left(\bar{X} > \mu_0 + z_\alpha \frac{\sigma}{\sqrt{n}}\right) &= P_\mu\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} > \frac{\sqrt{n}(\mu_0-\mu)}{\sigma} + z_\alpha\right) \\ &= 1 - \Phi\left(\frac{\sqrt{n}(\mu_0-\mu)}{\sigma} + z_\alpha\right) \end{aligned} \tag{49}$$

  is a strictly increasing function of μ.

- This implies that

$$\sup_{\mu \le \mu_0} \mathrm{E}_\mu[d^{\mathrm{LR}}(\boldsymbol{X})] = \mathrm{E}_{\mu_0}[d^{\mathrm{LR}}(\boldsymbol{X})] = \alpha. \tag{50}$$

Uniformly Most Powerful Test: Gaussian

- Furthermore, for any test $d(\boldsymbol{X})$ satisfying $\sup_{\mu \le \mu_0} \mathrm{E}_\mu[d(\boldsymbol{X})] \le \alpha$, it naturally follows that $\mathrm{E}_{\mu_0}[d(\boldsymbol{X})] \le \alpha$. Since d^{LR} is the UMP test for $H_0 : \mu = \mu_0$ vs. $H_1 : \mu > \mu_0$, we have:

$$\mathrm{E}_\mu[d^{\mathrm{LR}}(\boldsymbol{X})] \ge \mathrm{E}_\mu[d(\boldsymbol{X})] \quad \text{for any } \mu > \mu_0. \tag{51}$$

- Therefore, d^{LR} is the UMP test of level α for

$$H_0 : \mu \le \mu_0 \quad \text{vs.} \quad H_1 : \mu > \mu_0. \tag{52}$$

- Similarly, for testing $H_0 : \mu \ge \mu_0$ vs. $H_1 : \mu < \mu_0$,

$$d^{\mathrm{LR}}(\boldsymbol{X}) = I\left(\bar{X} < \mu_0 - z_\alpha \frac{\sigma}{\sqrt{n}}\right) \tag{53}$$

is the UMP test of level α.

Uniformly Most Powerful Test: Gaussian

Exercise 4

Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with a known μ and an unknown σ^2. Suppose we test $H_0 : \sigma^2 \leq \sigma_0^2$ vs. $H_1 : \sigma^2 > \sigma_0^2$. Find the uniformly most powerful test of level α.

Uniformly Most Powerful Test: Gaussian

Uniformly Most Powerful Test: Exponential Family

- Consider a single-parameter exponential family with PDF (or PMF):

$$f(x;\eta) = h(x)\exp\left(\eta T(x) - A(\eta)\right) \tag{54}$$

  where $\eta = \eta(\theta)$ is the natural parameter, $T(x)$ is a sufficient statistic, and $A(\eta)$ is the log-partition function.
- Suppose we test $H_0 : \theta \leq \theta_0$ vs. $H_1 : \theta > \theta_0$, and $\eta(\theta)$ is a strictly increasing function of θ. Then, the uniformly most powerful test of level α is given by:

$$d^{\text{UMP}}(\boldsymbol{X}) = \begin{cases} 1 & \text{if } \sum_{i=1}^n T(X_i) > c \\ \gamma & \text{if } \sum_{i=1}^n T(X_i) = c \\ 0 & \text{if } \sum_{i=1}^n T(X_i) < c \end{cases} \tag{55}$$

  where c and γ are determined by the significance level α.
- The uniformly most powerful test depends on the data only through the sufficient statistic $\sum_{i=1}^n T(X_i)$.

Uniformly Most Powerful Test: Exponential

Exercise 5

Let $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Exp}(\lambda)$ with an unknown λ. Suppose we test $H_0 : \lambda \leq \lambda_0$ vs. $H_1 : \lambda > \lambda_0$. Find the uniformly most powerful test of level α.

Non-existence of a Uniformly Most Powerful Test

- Unfortunately, a UMP test does not always exist, particularly for two-sided alternative hypotheses.
- For example, let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with an unknown mean μ and a known variance σ^2. Suppose we test:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0.$$

- If a UMP test $d^{\text{UMP}}(\boldsymbol{X})$ existed, then for any $\mu_- < \mu_0 < \mu_+$, it would have to be the most powerful test for both

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu = \mu_+ \tag{56}$$

and

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu = \mu_-. \tag{57}$$

- However, the rejection regions of the likelihood ratio tests for these two cases are in entirely opposite directions, which leads to a contradiction.

Uniformly Most Powerful Unbiased Test

- Popular remedies for this issue include (i) the uniformly most powerful unbiased test and (ii) the generalized likelihood ratio test. We will study the latter in detail in Section 9.4.
- For (i), instead of looking for a test that is most powerful among all possible tests, we restrict the class of competing tests.
- Consider a naive test $d^{\text{Naive}}(\boldsymbol{X})$ which ignores the observations $\boldsymbol{X}$ and simply rejects H_0 with a fixed probability α. Its power is given by $\mathrm{E}_\theta[d^{\text{Naive}}(\boldsymbol{X})] = \alpha$ for any $\theta \in \Omega_1$.
- In this context, when we test two hypotheses $H_0 : \theta \in \Omega_0$ vs. $H_1 : \theta \in \Omega_1$ where $\Omega_0 \cap \Omega_1 = \emptyset$, a test $d(\boldsymbol{X})$ is called an *unbiased test* of level α if it satisfies:

$$\sup_{\theta \in \Omega_0} \mathrm{E}_\theta[d(\boldsymbol{X})] \leq \alpha \text{ and } \inf_{\theta \in \Omega_1} \mathrm{E}_\theta[d(\boldsymbol{X})] \geq \alpha. \tag{58}$$

Uniformly Most Powerful Unbiased Test

Exercise 6

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with unknown μ and known σ^2. We test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$. Show that the standardized two-sided test

$$d^*(\boldsymbol{X}) = I\left(\frac{\sqrt{n}|\bar{X} - \mu_0|}{\sigma} > z_{\alpha/2}\right) \tag{59}$$

is unbiased.

Uniformly Most Powerful Unbiased Test

Uniformly Most Powerful Unbiased Test

- **Uniformly Most Powerful Unbiased Test**: Given a random sample $\mathbf{X} = (X_1, \ldots, X_n)^T$ from a parametric distribution with PDF $f_\theta(\boldsymbol{x})$ (or PMF $p_\theta(\boldsymbol{x})$ if discrete), suppose we test two hypotheses:

$$H_0 : \theta \in \Omega_0 \quad \text{vs.} \quad H_1 : \theta \in \Omega_1, \tag{60}$$

where $\Omega_0 \cap \Omega_1 = \emptyset$. A test $d^{\text{UMPU}}(\boldsymbol{X}) \in [0, 1]$ is defined as the *uniformly most powerful unbiased test* of level $0 < \alpha < 1$ if it satisfies the following conditions:

1. $\sup_{\theta \in \Omega_0} \mathrm{E}_\theta[d^{\text{UMPU}}(\boldsymbol{X})] \leq \alpha$ and $\inf_{\theta \in \Omega_1} \mathrm{E}_\theta[d^{\text{UMPU}}(\boldsymbol{X})] \geq \alpha$.
2. $\mathrm{E}_{\theta_1}[d^{\text{UMPU}}(\boldsymbol{X})] \geq \mathrm{E}_{\theta_1}[d(\boldsymbol{X})]$ for any $\theta_1 \in \Omega_1$ and any $d(\boldsymbol{X}) \in [0, 1]$ such that $\sup_{\theta \in \Omega_0} \mathrm{E}_\theta[d(\boldsymbol{X})] \leq \alpha$ and $\inf_{\theta \in \Omega_1} \mathrm{E}_\theta[d(\boldsymbol{X})] \geq \alpha$.

Uniformly Most Powerful Unbiased Test: Gaussian

Exercise 7

Let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with unknown μ and known σ^2. We test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$.

a) Let $\pi_d(\mu) = \mathrm{E}_\mu[d(\boldsymbol{X})]$ denote the power function of a test d. For an unbiased test d of level α, the continuity of $\pi_d(\mu)$ implies that $\pi_d(\mu_0) = \alpha$. By Fermat's theorem on stationary points[a], it follows that $\pi'_d(\mu_0) = 0$. Show that:

$$\pi'_d(\mu_0) = \mathrm{E}_{\mu_0}\left[d(\boldsymbol{X})\frac{n(\bar{X} - \mu_0)}{\sigma^2}\right]. \tag{61}$$

[a] When a differentiable function attains a local extremum at an interior point, its derivative at that point is zero.

Uniformly Most Powerful Unbiased Test: Gaussian

Uniformly Most Powerful Unbiased Test: Gaussian

b) For any test $d(\boldsymbol{X}) \in [0,1]$ and any constants k_1, k_2, show that the power $\mathrm{E}_{\mu_1}[d(\boldsymbol{X})]$ at a given $\mu_1 \neq \mu_0$ can be expressed as:

$$\mathrm{E}_{\mu_1}[d(\boldsymbol{X})] = \mathrm{E}_{\mu_0}\left[d(\boldsymbol{X})\left(\frac{f_{\mu_1}(\boldsymbol{X})}{f_{\mu_0}(\boldsymbol{X})} - k_1 - k_2\frac{n(\bar{X}-\mu_0)}{\sigma^2}\right)\right]$$
$$+ k_1\,\mathrm{E}_{\mu_0}[d(\boldsymbol{X})] + k_2\,\mathrm{E}_{\mu_0}\left[d(\boldsymbol{X})\frac{n(\bar{X}-\mu_0)}{\sigma^2}\right].$$

Uniformly Most Powerful Unbiased Test: Gaussian

Uniformly Most Powerful Unbiased Test: Gaussian

c) Let k_1, k_2 be constants, and define:

$$d^*_{k_1,k_2}(\boldsymbol{X}) = I\left(\frac{f_{\mu_1}(\boldsymbol{X})}{f_{\mu_0}(\boldsymbol{X})} > k_1 + k_2\frac{n(\bar{X}-\mu_0)}{\sigma^2}\right). \tag{62}$$

Show that for any unbiased test $d(\boldsymbol{X})$ of level α and any given $\mu_1 \neq \mu_0$, the following holds:

$$\begin{aligned} \mathrm{E}_{\mu_1}[d(\boldsymbol{X})] \leq \mathrm{E}_{\mu_1}[d^*_{k_1,k_2}(\boldsymbol{X})] + k_1\left(\alpha - \mathrm{E}_{\mu_0}[d^*_{k_1,k_2}(\boldsymbol{X})]\right) \\ - k_2\,\mathrm{E}_{\mu_0}\left[d^*_{k_1,k_2}(\boldsymbol{X})\frac{n(\bar{X}-\mu_0)}{\sigma^2}\right]. \end{aligned} \tag{63}$$

Uniformly Most Powerful Unbiased Test: Gaussian

Uniformly Most Powerful Unbiased Test: Gaussian

- Note that Equation (61) can be generalized as:

$$\pi'_d(\theta_0) = \mathrm{E}_{\theta_0}\left[d(\boldsymbol{X})\frac{\dot{p}_{\theta_0}(\boldsymbol{X})}{p_{\theta_0}(\boldsymbol{X})}\right], \tag{64}$$

  where $\dot{p}_{\theta_0}(\boldsymbol{X}) = \frac{\partial}{\partial\theta}p_\theta(\boldsymbol{X})\Big|_{\theta=\theta_0}$.
- Based on the previous exercise, if there exist constants k_1 and k_2 such that $d^*_{k_1,k_2}(\boldsymbol{X})$ is an unbiased test, then it is the uniformly most powerful unbiased test.
- With appropriately chosen k_1 and k_2, we can recover the standard two-sided test as follows:[5]

$$d^*(\boldsymbol{X}) = I\left(\frac{\sqrt{n}|\bar{X}-\mu_0|}{\sigma} > z_{\alpha/2}\right). \tag{65}$$

[5] Specifically, this occurs when $k_1 = e^B\cosh(Ac)$ and $k_2 = \frac{e^B\sinh(Ac)}{c}$, where $A = \frac{n(\mu_1-\mu_0)}{\sigma^2}$, $B = -\frac{n(\mu_1-\mu_0)^2}{2\sigma^2}$, and $c = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$.

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The Duality of Confidence Intervals and Hypothesis Tests

- We can construct confidence intervals from the acceptance regions of hypothesis tests, and vice versa.
- For example, let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with unknown μ and known σ^2. Suppose we test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$ using the standard two-sided test.
- First, the acceptance region depends on μ_0:

$$A(\mu_0) := \left\{ \boldsymbol{X} : \frac{\sqrt{n}|\bar{X} - \mu_0|}{\sigma} \leq z_{\alpha/2} \right\}. \tag{66}$$

Given $A(\mu_0)$, let

$$C(\boldsymbol{X}) := \{\mu_0 : \boldsymbol{X} \in A(\mu_0)\} = \left[\bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \right]. \tag{67}$$

Then, $C(\boldsymbol{X})$ is a $100(1-\alpha)\%$ confidence interval for μ.

The Duality of Confidence Intervals and Hypothesis Tests

- Conversely, given a confidence region $C(\boldsymbol{X})$, the corresponding acceptance region for $H_0 : \mu = \mu_0$ can be expressed as:

$$A(\mu_0) = \left\{ \boldsymbol{X} : \mu_0 \in C(\boldsymbol{X}) \right\}. \tag{68}$$

- This duality holds in general for

$$A(\theta_0) = \left\{ \boldsymbol{X} : \theta_0 \in C(\boldsymbol{X}) \right\} \quad \text{and} \quad C(\boldsymbol{X}) = \left\{ \theta_0 : \boldsymbol{X} \in A(\theta_0) \right\}. \tag{69}$$

See Theorems 9.3.A and 9.3.B for details.

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Generalized Likelihood Ratio Tests

- The Neyman-Pearson Lemma justifies the use of likelihood ratio tests for simple hypotheses. We have also studied its extensions to composite hypotheses, including UMP tests.
- However, these extensions are not always applicable depending on the specific form of the hypotheses. For example, deriving a UMP (unbiased) test is usually difficult.
- In this section, we introduce the *generalized likelihood ratio (GLR) test*, which leverages MLEs to construct practical testing procedures for general composite hypotheses.

Generalized Likelihood Ratio Tests

- Suppose we have data $X_i \overset{\text{i.i.d.}}{\sim} f_\theta(x)$ with an unknown parameter θ, and we test the hypotheses:

$$H_0 : \theta \in \Omega_0 \quad \text{vs.} \quad H_1 : \theta \in \Omega_1 \tag{70}$$

  where $\Omega_0 \cap \Omega_1 = \emptyset$ and $\Omega := \Omega_0 \cup \Omega_1$.

- The GLR test utilizes:

$$\Lambda^*(\boldsymbol{X}) = \frac{\max_{\theta \in \Omega_0} L(\theta; \boldsymbol{X})}{\max_{\theta \in \Omega_1} L(\theta; \boldsymbol{X})}. \tag{71}$$

  Smaller values of Λ^* indicate that the data $\boldsymbol{X}$ more strongly support H_1 over H_0.

Generalized Likelihood Ratio Tests

- It is more convenient to use the ratio over the entire parameter space Ω:

$$\Lambda(\boldsymbol{X}) = \frac{\max_{\theta \in \Omega_0} L(\theta; \boldsymbol{X})}{\max_{\theta \in \Omega} L(\theta; \boldsymbol{X})}. \tag{72}$$

- Note that, for any λ_0,

$$\Lambda^*(\boldsymbol{X}) \leq \lambda_0 \iff \Lambda(\boldsymbol{X}) = \min\{1, \Lambda^*(\boldsymbol{X})\} \leq \min\{1, \lambda_0\}. \tag{73}$$

- Specifically, the GLR test rejects H_0 for small values of $\Lambda(\boldsymbol{X})$. For a given threshold $0 < \lambda_0 \leq 1$, the test function is:

$$d^{\text{GLR}}(\boldsymbol{X}) = \begin{cases} 1 & \text{if} \quad \Lambda(\boldsymbol{X}) < \lambda_0 \\ \gamma & \text{if} \quad \Lambda(\boldsymbol{X}) = \lambda_0 \\ 0 & \text{if} \quad \Lambda(\boldsymbol{X}) > \lambda_0. \end{cases} \tag{74}$$

As in the likelihood ratio test, we can typically ignore the $\Lambda(\boldsymbol{X}) = \lambda_0$ case for continuous distributions.

Generalized Likelihood Ratio Tests: Gaussian

Exercise 8

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with unknown μ and known σ^2. We test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$. Show that the standardized two-sided test given by

$$d^{\text{GLR}}(\boldsymbol{X}) = I\left(\frac{\sqrt{n}|\bar{X} - \mu_0|}{\sigma} > z_{\alpha/2}\right) \tag{75}$$

is the GLR test of level α.[a]

[a] As we showed previously, it is the UMP unbiased test.

Generalized Likelihood Ratio Tests: Gaussian

Generalized Likelihood Ratio Tests: Gaussian

Exercise 9

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with an unknown parameter vector $\theta = (\mu, \sigma^2)^T$, where $n \geq 2$. We test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$.

a) Show that

$$d^{\text{GLR}}(\boldsymbol{X}) = I\left(\frac{\sqrt{n}|\bar{X} - \mu_0|}{S} > t_{\alpha/2}(n-1)\right) \tag{76}$$

is the GLR test of level α.

Generalized Likelihood Ratio Tests: Gaussian

Generalized Likelihood Ratio Tests: Gaussian

b) Let $\pi_{d^{\text{GLR}}}(\theta) = \mathrm{E}_\theta[d^{\text{GLR}}(\boldsymbol{X})]$ be the power function of the GLR test. Show that:

$$\pi_{d^{\text{GLR}}}(\theta) = \mathrm{E}_\theta\left[1 - \Phi\left(\frac{S}{\sigma}t_{\alpha/2}(n-1) - \frac{\sqrt{n}(\mu-\mu_0)}{\sigma}\right) + \Phi\left(-\frac{S}{\sigma}t_{\alpha/2}(n-1) - \frac{\sqrt{n}(\mu-\mu_0)}{\sigma}\right)\right]. \tag{77}$$

Generalized Likelihood Ratio Tests: Gaussian

Generalized Likelihood Ratio Tests: Gaussian

c) Show that the GLR test is unbiased.[a]

[a] It is also the UMP unbiased test.

Generalized Likelihood Ratio Tests: Gaussian

Generalized Likelihood Ratio Tests: Gaussian

- In the previous exercise, under H_1,

$$T := \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} = \frac{Z + \delta}{\sqrt{V/(n-1)}}, \tag{78}$$

  where $Z = \sqrt{n}(\bar{X} - \mu)/\sigma$, $V = (n-1)S^2/\sigma^2$, and $\delta = \sqrt{n}(\mu - \mu_0)/\sigma$. Note that $Z \sim \mathcal{N}(0,1)$, $V \sim \chi^2(n-1)$, and $Z \perp\!\!\!\perp V$.
- The distribution of T is referred to as the *noncentral t distribution* with $(n-1)$ degrees of freedom and noncentrality parameter δ.
- When $\delta = 0$, i.e., $\mu = \mu_0$, the distribution of T reduces to the t distribution with $(n-1)$ degrees of freedom.

Generalized Likelihood Ratio Tests: Gaussian

Exercise 10

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with an unknown parameter vector $\theta = (\mu, \sigma^2)^T$, where $n \geq 2$. We test $H_0 : \sigma^2 = \sigma_0^2 (> 0)$ vs. $H_1 : \sigma^2 \neq \sigma_0^2$.

a) Show that, for any $0 < \lambda_0 \leq 1$,

$$\Lambda(\boldsymbol{X}) < \lambda_0 \iff \frac{(n-1)S^2}{n\sigma_0^2} < c_L \quad \text{or} \quad \frac{(n-1)S^2}{n\sigma_0^2} > c_U, \tag{79}$$

where the constants c_L and c_U satisfy $c_L \leq c_U$ and

$$\log \lambda_0 = \frac{n}{2}\left(\log c_L - c_L + 1\right) = \frac{n}{2}\left(\log c_U - c_U + 1\right).$$

Generalized Likelihood Ratio Tests: Gaussian

Generalized Likelihood Ratio Tests: Gaussian

b) Let $V \sim \chi^2(n-1)$. Show that the GLR test of level α is given by:

$$d^{\mathrm{GLR}}(\boldsymbol{X}) = I\left(\frac{(n-1)S^2}{n\sigma_0^2} < c_L \quad \text{or} \quad \frac{(n-1)S^2}{n\sigma_0^2} > c_U\right), \tag{80}$$

where c_L and c_U satisfy $c_L \leq c_U$, $\log c_L - c_L = \log c_U - c_U$, and $1-\alpha = F_V(nc_U) - F_V(nc_L)$.

Generalized Likelihood Ratio Tests: Gaussian

Generalized Likelihood Ratio Tests: Gaussian

c) Let $\pi_{d^{\text{GLR}}}(\theta) = \mathrm{E}_\theta[d^{\text{GLR}}(\boldsymbol{X})]$ be the power function of the GLR test. For $V \sim \chi^2(n-1)$, show that:

$$\pi_{d^{\text{GLR}}}(\theta) = 1 - F_V\left(\frac{n\sigma_0^2 c_U}{\sigma^2}\right) + F_V\left(\frac{n\sigma_0^2 c_L}{\sigma^2}\right), \tag{81}$$

where c_L and c_U satisfy $c_L \leq c_U$, $\log c_L - c_L = \log c_U - c_U$, and $1 - \alpha = F_V(nc_U) - F_V(nc_L)$. Note that the power function does not depend on μ.

Generalized Likelihood Ratio Tests: Gaussian

Generalized Likelihood Ratio Tests: Gaussian

d) For these hypotheses, any unbiased test d should satisfy $\pi'_d((\mu, \sigma_0^2)^T) = 0$. Show that $\pi'_{d^{\mathrm{GLR}}}((\mu, \sigma_0^2)^T) \neq 0$.

Generalized Likelihood Ratio Tests: Gaussian

Generalized Likelihood Ratio Tests: Gaussian

- Note that the UMP unbiased test exists and is given by:

$$d^{\text{UMPU}}(\boldsymbol{X}) = I\left(\frac{(n-1)S^2}{n\sigma_0^2} < c_L' \quad \text{or} \quad \frac{(n-1)S^2}{n\sigma_0^2} > c_U'\right), \tag{82}$$

  where $V_\nu \sim \chi^2(\nu)$, and the constants c_L' and c_U' satisfy $c_L' \leq c_U'$ and $F_{V_{n-1}}(nc_U') - F_{V_{n-1}}(nc_L') = F_{V_{n+1}}(nc_U') - F_{V_{n+1}}(nc_L') = 1 - \alpha$.
- Based on the previous exercise, the GLR test is **biased**:

$$d^{\text{GLR}}(\boldsymbol{X}) = I\left(\frac{(n-1)S^2}{n\sigma_0^2} < c_L \quad \text{or} \quad \frac{(n-1)S^2}{n\sigma_0^2} > c_U\right), \tag{83}$$

  where the constants c_L and c_U satisfy $c_L \leq c_U$, $\log c_L - c_L = \log c_U - c_U$, and $F_{V_{n-1}}(nc_U) - F_{V_{n-1}}(nc_L) = 1 - \alpha$.
- However, $d^{\text{UMPU}}(\boldsymbol{X}) \approx d^{\text{GLR}}(\boldsymbol{X})$ when n is large.

Generalized Likelihood Ratio Tests: Exponential

Exercise 11

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{Exp}(\theta)$ with an unknown parameter θ. We test $H_0 : \theta = \theta_0 (> 0)$ vs. $H_1 : \theta \neq \theta_0$.

a) Show that, for any $0 < \lambda_0 \leq 1$,

$$\Lambda(\boldsymbol{X}) < \lambda_0 \iff \theta_0 \bar{X} < c_L \quad \text{or} \quad \theta_0 \bar{X} > c_U, \tag{84}$$

where the constants c_L and c_U satisfy $c_L \leq c_U$ and

$$\log \lambda_0 = n\left(\log c_L - c_L + 1\right) = n\left(\log c_U - c_U + 1\right) \tag{85}$$

Generalized Likelihood Ratio Tests: Exponential

Generalized Likelihood Ratio Tests: Exponential

b) Let $W \sim \chi^2(2n)$. Show that the GLR test of level α is given by:

$$d^{\mathrm{GLR}}(\boldsymbol{X}) = I\left(\theta_0 \bar{X} < c_L \quad \text{or} \quad \theta_0 \bar{X} > c_U\right), \tag{86}$$

where c_L and c_U satisfy $c_L \leq c_U$, $\log c_L - c_L = \log c_U - c_U$, and $1 - \alpha = F_W(2nc_U) - F_W(2nc_L)$.

Generalized Likelihood Ratio Tests: Exponential

Generalized Likelihood Ratio Tests: Exponential

c) Let $\pi_{d^{\text{GLR}}}(\theta) = \mathrm{E}_\theta[d^{\text{GLR}}(\boldsymbol{X})]$ be the power function of the GLR test. For $W \sim \chi^2(2n)$, show that:

$$\pi_{d^{\text{GLR}}}(\theta) = 1 - F_W\left(\frac{2n\theta c_U}{\theta_0}\right) + F_W\left(\frac{2n\theta c_L}{\theta_0}\right), \tag{87}$$

where c_L and c_U satisfy $c_L \leq c_U$, $\log c_L - c_L = \log c_U - c_U$, and $1 - \alpha = F_W(2nc_U) - F_W(2nc_L)$.

Generalized Likelihood Ratio Tests: Exponential

d) Show that the GLR test is unbiased.[a]

[a] This two-sided test is also the UMP unbiased test.

Approximation of GLR Tests

- It is generally challenging to derive the exact sampling distributions of GLR test statistics. However, under some regularity conditions, we can derive their asymptotic distribution.
- Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} f_\theta(x)$ with an unknown θ. When we test $H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta \neq \theta_0$, the GLR statistic is given by:

$$\Lambda(\boldsymbol{X}) = \frac{L(\theta_0; \boldsymbol{X})}{L(\hat{\theta}^{\text{MLE}}; \boldsymbol{X})}. \tag{88}$$

- Under some regularity conditions for the asymptotic normality of MLEs, $l_n'(\hat{\theta}_n^{\text{MLE}}) = 0$ with high probability, implying:

$$-2 \log \Lambda(\boldsymbol{X}) = 2 \left(l_n(\hat{\theta}_n^{\text{MLE}}) - l_n(\theta_0) \right) = -l_n''(\tilde{\theta}_n) \left(\hat{\theta}_n^{\text{MLE}} - \theta_0 \right)^2, \tag{89}$$

where $\tilde{\theta}_n$ lies between θ_0 and $\hat{\theta}_n^{\text{MLE}}$.

Approximation of GLR Tests

- Recall that, in Chapter 8, we derived the asymptotic properties of MLEs. Under $H_0 : \theta = \theta_0$ and some regularity conditions:

$$\sqrt{n}\left(\hat{\theta}_n^{\text{MLE}} - \theta_0\right) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}_1(\theta_0)^{-1}), \tag{90}$$

  where $\mathcal{I}_n(\theta) = \text{Var}_\theta[s_n(\theta)]$ and $s_n(\theta) = l_n'(\theta)$ represent the Fisher information and the score function, respectively.
- Furthermore, by the consistency of MLE and the Law of Large Numbers:

$$\frac{1}{n} l_n''(\tilde{\theta}_n) \xrightarrow{p} -\mathcal{I}_1(\theta_0). \tag{91}$$

- Combining these results implies that, under H_0:

$$-2\log\Lambda(\boldsymbol{X}) \xrightarrow{d} V, \quad \text{where} \quad V \sim \chi^2(1). \tag{92}$$

  Therefore, when n is large, the GLR test of level α can be approximated as:

$$d^{\text{GLR}}(\boldsymbol{X}) \approx I\left(-2\log\Lambda(\boldsymbol{X}) > \chi^2_\alpha(1)\right). \tag{93}$$

Approximation of GLR Tests

- For a k-dimensional parameter space $\Omega \subseteq \mathbb{R}^k$, consider testing a general hypothesis $H_0 : \theta \in \Omega_0$ vs. $H_1 : \theta \in \Omega \setminus \Omega_0$.
- If H_0 is a simple hypothesis ($H_0 : \theta = \theta_0$), the multivariate Taylor expansion yields:

$$-2 \log \Lambda(\boldsymbol{X}) = -\left(\hat{\theta}_n^{\text{MLE}} - \theta_0\right)^T \left(\nabla^2 l_n(\tilde{\theta}_n)\right) \left(\hat{\theta}_n^{\text{MLE}} - \theta_0\right) \xrightarrow{d} V_k \sim \chi^2(k). \tag{94}$$

- **<u>Wilks' Theorem</u>**: For a general composite hypothesis, under H_0 and suitable regularity conditions, as $n \to \infty$:

$$-2 \log \Lambda(\boldsymbol{X}) \xrightarrow{d} V, \quad \text{where} \quad V \sim \chi^2(\dim(\Omega) - \dim(\Omega_0)), \tag{95}$$

where $\dim(\cdot)$ represents the dimensionality of the parameter space.

Approximation of GLR Tests: Gaussian

- For example, let $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with an unknown $\theta = (\mu, \sigma^2)^T$, where $n \geq 2$. When we test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$, the GLR test statistic is given by:

$$-2 \log \Lambda(\boldsymbol{X}) = n \log \left(1 + \frac{T^2}{n-1} \right), \tag{96}$$

  where $T = \sqrt{n}(\bar{X} - \mu_0)/S$. Here, $T \sim t(n-1)$ under H_0.
- By the Taylor expansion of $\log(1+x)$ at $x = 0$:

$$-2 \log \Lambda(\boldsymbol{X}) = n \left(\frac{T^2}{n-1} - \frac{1}{2(1+\tilde{\xi})^2} \frac{T^4}{(n-1)^2} \right), \tag{97}$$

  where $\tilde{\xi}$ lies between 0 and $\frac{T^2}{n-1}$.

Approximation of GLR Tests: Gaussian

- Since $T \xrightarrow{d} Z \sim \mathcal{N}(0,1)$ under H_0, and the remainder term converges to 0 in probability,

$$-2\log\Lambda(\boldsymbol{X}) \xrightarrow{d} Z^2 \sim \chi^2(1), \tag{98}$$

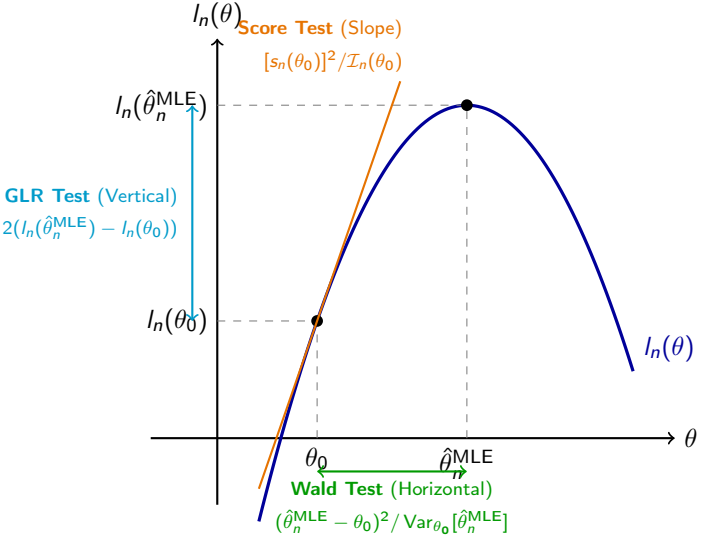
  which matches Wilks' Theorem. Note that $\dim(\Omega) = 2$ and $\dim(\Omega_0) = 1$.
- Therefore, the GLR test of level α is asymptotically given by:

$$d^{\mathrm{GLR}}(\boldsymbol{X}) \approx I\left(\frac{n(\bar{X}-\mu_0)^2}{S^2} > \chi^2_\alpha(1)\right). \tag{99}$$

- Note that, under H_1, $(\bar{X}-\mu_0)^2/S^2 \xrightarrow{p} (\mu-\mu_0)^2/\sigma^2 > 0$. Therefore, the GLR test is *consistent*, i.e.,

$$\lim_{n\to\infty} \pi_{d^{\mathrm{GLR}}}(\theta) = 1.$$

The Trinity of Hypothesis Tests



The Trinity of Hypothesis Tests

- **Wald Test Statistic**:[6] The *Wald test statistic* is defined as the normalized (squared) horizontal distance between $\hat{\theta}_n^{\text{MLE}}$ and θ_0:

$$W(\boldsymbol{X}) = (\hat{\theta}_n^{\text{MLE}} - \theta_0)^T \left(\text{Var}_{\theta_0}[\hat{\theta}_n^{\text{MLE}}]\right)^{-1} (\hat{\theta}_n^{\text{MLE}} - \theta_0). \tag{100}$$

- **Rao's Score Test Statistic**: *Rao's score test statistic* (or Lagrange Multiplier test statistic) is defined as the normalized squared slope (gradient) of the log-likelihood at θ_0:

$$S(\boldsymbol{X}) = s_n(\theta_0)^T \mathcal{I}_n(\theta_0)^{-1} s_n(\theta_0). \tag{101}$$

- **The Trinity of Hypothesis Tests**: Under $H_0 : \theta = \theta_0$, all three foundational tests, GLR, Wald, and Rao's score, are asymptotically equivalent.

[6] The Wald statistic is also commonly defined using the observed Fisher information: $W(\boldsymbol{X}) = (\hat{\theta}_n^{\text{MLE}} - \theta_0)^T \mathcal{I}_n(\hat{\theta}_n^{\text{MLE}})(\hat{\theta}_n^{\text{MLE}} - \theta_0)$.

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Likelihood Ratio Tests for the Multinomial Distribution

- <u>**Multinomial Random Vector**</u>: A random vector $\boldsymbol{X}$ is said to follow the *multinomial distribution*, denoted by $\boldsymbol{X} \sim \mathrm{Multin}(n, \boldsymbol{p})$, where $n \in \mathbb{N}$, $\boldsymbol{p} = (p_1, \ldots, p_d)^T$, $p_i \geq 0$, and $\sum_{i=1}^{d} p_i = 1$, if its PMF is given by:

$$p(\boldsymbol{x}) = \begin{cases} \binom{n}{x_1, x_2, \ldots, x_d} p_1^{x_1} p_2^{x_2} \cdots p_d^{x_d} & \text{if } x_i \in \{0, 1, \ldots, n\} \text{ and } \sum_{i=1}^{d} x_i = n, \\ 0 & \text{otherwise.} \end{cases} \tag{102}$$

Likelihood Ratio Tests for the Multinomial Distribution

- Let $\boldsymbol{p} = (p_1, \ldots, p_{14})^T \in \{\boldsymbol{p} : p_j \geq 0, \sum_{j=1}^{14} p_j = 1\}$.

Outcome	1st □	2nd □	3rd □	4th □	5th □	6th □	1st ⬡
Probability	p_1	p_2	p_3	p_4	p_5	p_6	p_7

Outcome	2nd ⬡	3rd ⬡	4th ⬡	5th ⬡	6th ⬡	7th ⬡	8th ⬡
Probability	p_8	p_9	p_{10}	p_{11}	p_{12}	p_{13}	p_{14}

- Suppose we test whether all faces are equally likely:

$$H_0 : \boldsymbol{p} = (1/14, \ldots, 1/14)^T \quad \text{vs.} \quad H_1 : \boldsymbol{p} \neq (1/14, \ldots, 1/14)^T. \tag{103}$$

Likelihood Ratio Tests for the Multinomial Distribution

Exercise 12

Let $\boldsymbol{X}_i := (X_{i1}, \ldots, X_{id})^T \overset{\text{i.i.d.}}{\sim} \mathrm{Multin}(1, \boldsymbol{p})$ where $i = 1, \ldots, n$. Show that:

$$\hat{\boldsymbol{p}}^{\text{MLE}} = \left(\bar{X}_{\cdot 1}, \ldots, \bar{X}_{\cdot d}\right)^T, \tag{104}$$

where $\bar{X}_{\cdot j} = n^{-1} \sum_{i=1}^{n} X_{ij}$ and $j = 1, \ldots, d$.

Likelihood Ratio Tests for the Multinomial Distribution

Likelihood Ratio Tests for the Multinomial Distribution

- Suppose we test $H_0 : \boldsymbol{p} = \boldsymbol{p}_0$ vs. $H_1 : \boldsymbol{p} \neq \boldsymbol{p}_0$, where $\boldsymbol{p}_0 := (p_{01}, \ldots, p_{0d})^T$.
- For n i.i.d. samples $\boldsymbol{X}_i := (X_{i1}, \ldots, X_{id})^T \sim \mathrm{Multin}(1, \boldsymbol{p})$,

$$\Lambda((\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)) = \frac{\prod_{i=1}^n p_{01}^{X_{i1}} \cdots p_{0d}^{X_{id}}}{\prod_{i=1}^n \left(\hat{p}_1^{\mathrm{MLE}}\right)^{X_{i1}} \cdots \left(\hat{p}_d^{\mathrm{MLE}}\right)^{X_{id}}} = \prod_{j=1}^d \left(\frac{p_{0j}}{\hat{p}_j^{\mathrm{MLE}}}\right)^{n\bar{X}_{\cdot j}}. \tag{105}$$

- Therefore, the GLR test statistic is given by:[7]

$$-2 \log \Lambda((\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)) = 2 \sum_{j=1}^d O_j \log\left(\frac{O_j}{E_j}\right), \tag{106}$$

where $O_j = n\hat{p}_j^{\mathrm{MLE}}$ and $E_j = np_{0j}$ represent the observed and expected counts.

[7] When $O_j = 0$, or equivalently $\hat{p}_j^{\mathrm{MLE}} = 0$, we define $O_j \log O_j = 0$.

Likelihood Ratio Tests for the Multinomial Distribution

- Under $H_0 : \boldsymbol{p} = \boldsymbol{p}_0$, by Wilks' Theorem, the GLR test statistic asymptotically follows $\chi^2(d-1)$, since $\dim(\Omega) = d - 1$ and $\dim(\Omega_0) = 0$.
- Wang et al. (2005) observed experimental results for the wooden die ($n = 7000$):

Outcome	1st □	2nd □	3rd □	4th □	5th □	6th □	1st ⬡
Observed Counts	523	533	477	525	518	492	456

Outcome	2nd ⬡	3rd ⬡	4th ⬡	5th ⬡	6th ⬡	7th ⬡	8th ⬡
Observed Counts	512	506	488	505	487	482	496

- Based on this, $E_j = 7000 \times (1/14) = 500$ for all j. The GLR test statistic is

$$-2\log\Lambda = 2\sum_{j=1}^{14} O_j \log\left(\frac{O_j}{E_j}\right) = 2\left(523\log\frac{523}{500} + \cdots + 496\log\frac{496}{500}\right) = 11.965. \tag{107}$$

- Since $11.965 < \chi^2_{0.05}(13) \approx 22.36$, the GLR test fails to reject H_0 at $\alpha = 0.05$.

Likelihood Ratio Tests for the Multinomial Distribution

- In this case, both the Wald test statistic and Rao's score test statistic are identical, and are known as *Pearson's* χ^2 *statistic*:

$$W((\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)) = S((\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)) = \sum_{j=1}^{d} \frac{(O_j - E_j)^2}{E_j}. \tag{108}$$

- Computing this yields:

$$\frac{(523-500)^2}{500} + \frac{(533-500)^2}{500} + \cdots + \frac{(496-500)^2}{500} = 11.908.$$

Likelihood Ratio Tests for the Multinomial Distribution

- Suppose we test whether faces of the same shape have identical probabilities:

$$\begin{aligned} & H_0 : p_1 = \cdots = p_6 \quad \text{and} \quad p_7 = \cdots = p_{14} \\ \iff & H_0 : \boldsymbol{p} = (\underbrace{p/6, \ldots, p/6}_{6}, \underbrace{q/8, \ldots, q/8}_{8})^T \quad \text{vs.} \quad H_1 : \text{Not } H_0, \end{aligned} \tag{109}$$

  where $q = 1 - p$.
- Under H_0, the total log-likelihood becomes:

$$\begin{aligned} & \ell((\underbrace{p/6, \ldots, p/6}_{6}, \underbrace{q/8, \ldots, q/8}_{8})^T) \\ & = \left(\sum_{j=1}^{6} O_j \right) \log p + \left(n - \sum_{j=1}^{6} O_j \right) \log q + C(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n), \end{aligned} \tag{110}$$

  which matches the log-likelihood of a binomial random variable $\sum_{j=1}^{6} O_j \sim \mathrm{Bin}(n, p)$ up to constant addition.

Likelihood Ratio Tests for the Multinomial Distribution

- Therefore, the MLE for p under H_0 is given by:

$$\hat{p}^{\Omega_0} = \frac{\sum_{j=1}^{6} O_j}{n}. \tag{111}$$

- The GLR test statistic is then:

$$-2\log\Lambda((\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)) = 2\sum_{j=1}^{14} O_j \log\left(\frac{O_j}{E_j}\right), \tag{112}$$

where O_j represents the observed counts, and the expected counts under H_0 are:

$$E_j = \begin{cases} \frac{n\hat{p}^{\Omega_0}}{6} = \frac{1}{6}\sum_{k=1}^{6} O_k = 511.33 & j = 1, \ldots, 6 \\ \frac{n\hat{q}^{\Omega_0}}{8} = \frac{1}{8}\sum_{k=7}^{14} O_k = 491.50 & j = 7, \ldots, 14 \end{cases} \tag{113}$$

Likelihood Ratio Tests for the Multinomial Distribution

- Under H_0, by Wilks' Theorem, the GLR test statistic asymptotically follows $\chi^2(12)$, since $\dim(\Omega) = 13$ and $\dim(\Omega_0) = 1$.
- The GLR test statistic is calculated as:

$$-2\log\Lambda = 2\left(\sum_{j=1}^{6} O_j \log\left(\frac{O_j}{511.33}\right) + \sum_{j=7}^{14} O_j \log\left(\frac{O_j}{491.50}\right)\right) = 9.272. \tag{114}$$

- Since $9.272 < \chi^2_{0.05}(12) \approx 21.03$, the GLR test fails to reject H_0 at $\alpha = 0.05$.
- Pearson's χ^2 statistic is given by:

$$\sum_{j=1}^{14} \frac{(O_j - E_j)^2}{E_j} = \frac{(523 - 511.33)^2}{511.33} + \cdots + \frac{(496 - 491.50)^2}{491.50} \approx 9.181. \tag{115}$$

Practice Problem I

1. Let $X_1, \ldots, X_n$ be univariate continuous random samples with PDF $f(x \mid H_j)$ under hypothesis H_j for $j = 0, 1$. Let $R(\phi)$ denote the *Bayesian average error rate* of a test $\phi(\boldsymbol{X})$ where $\boldsymbol{X} = (X_1, \ldots, X_n)^T$, defined as:

$$R(\phi) = P(H_0)\,\mathrm{E}[\phi(\boldsymbol{X}) \mid H_0] + P(H_1)\,\mathrm{E}[1 - \phi(\boldsymbol{X}) \mid H_1]. \tag{116}$$

   a) For any test ϕ, show that:

$$R(\phi) = \int_{\mathbb{R}^n} \left(P(H_1) \prod_{i=1}^{n} f(x_i \mid H_1) + \phi(\boldsymbol{x}) \left(P(H_0) \prod_{i=1}^{n} f(x_i \mid H_0) - P(H_1) \prod_{i=1}^{n} f(x_i \mid H_1) \right) \right) d\boldsymbol{x}. \tag{117}$$

Practice Problem II

b) Consider the test ϕ^π given by:

$$\phi^\pi(\boldsymbol{x}) = \begin{cases} 1, & P(H_0 \mid \boldsymbol{x}) \leq P(H_1 \mid \boldsymbol{x}) \\ 0, & P(H_0 \mid \boldsymbol{x}) > P(H_1 \mid \boldsymbol{x}). \end{cases} \tag{118}$$

Show that $R(\phi^\pi) \leq R(\phi)$ for any test ϕ.

2. Let $X_1, \ldots, X_n$ be i.i.d. observations from a Poisson distribution with unknown parameter λ. Suppose we test two competing simple hypotheses:

$$H_0 : \lambda = \lambda_0 \quad \text{vs.} \quad H_1 : \lambda = \lambda_1, \tag{119}$$

where $\lambda_0 > \lambda_1$.

a) Assuming n is sufficiently large, apply the CLT to derive the likelihood ratio test at a significance level α, where $0 < \alpha < 1$.

b) Based on the test derived in part a), derive the power of the likelihood ratio test.

Practice Problem III

c) Determine the minimum sample size n required to guarantee a target power of at least $1 - \beta$.

3. Let $X_1, \ldots, X_n$ be i.i.d. observations from a Gamma distribution with $\theta = (r, \lambda)^T$ where r is known while λ is unknown. Suppose we test two competing simple hypotheses:

$$H_0 : \theta = \theta_0 = (r, \lambda_0)^T \quad \text{vs.} \quad H_1 : \theta = \theta_1 = (r, \lambda_1)^T, \tag{120}$$

where $\lambda_0 < \lambda_1$.

a) Assuming n is sufficiently large, apply the CLT to derive the likelihood ratio test at a significance level of α, where $0 < \alpha < 1$.
b) Based on the test derived in part a), derive the power of the likelihood ratio test.
c) Determine the minimum sample size n required to guarantee a target power of at least $1 - \beta$.

Practice Problem IV

4. The PDF of a *Weibull* with $\theta = (k, \lambda)^T$ where $k > 0$ and $\lambda > 0$ is given by:

$$f(x;\theta) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} \exp\left(-\left(\frac{x}{\lambda}\right)^{k}\right), \quad x \geq 0. \tag{121}$$

Let $X_1, \ldots, X_n$ be i.i.d. observations from a Weibull distribution with a known k and an unknown λ. Suppose we test the following simple hypotheses:

$$H_0 : \theta = \theta_0 = (k, \lambda_0)^T \quad \text{vs.} \quad H_1 : \theta = \theta_1 = (k, \lambda_1)^T, \tag{122}$$

where $\lambda_0 < \lambda_1$.

a) Assuming n is sufficiently large, apply the CLT to derive the likelihood ratio test at a significance level of α, where $0 < \alpha < 1$.
b) Based on the test derived in part a), derive the power of the likelihood ratio test.

Practice Problem V

c) Determine the minimum sample size n required to guarantee a target power of at least $1 - \beta$.

5. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{U}(0, \theta)$. Find the UMP test of level α for testing $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$ for a given $\theta_0 > 0$.

6. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with an unknown parameter vector $\theta = (\mu, \sigma^2)^T$, where $n \geq 2$. We test $H_0 : \mu \leq \mu_0$ vs. $H_1 : \mu > \mu_0$. Show that

$$d^{\text{GLR}}(\boldsymbol{X}) = I\left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{S} > t_\alpha(n-1)\right) \tag{123}$$

is the GLR test of level α.

Practice Problem VI

7. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with an unknown parameter vector $\theta = (\mu, \sigma^2)^T$, where $n \geq 2$. We test $H_0 : \sigma \leq \sigma_0 (> 0)$ vs. $H_1 : \sigma > \sigma_0$. Show that

$$d^{\text{GLR}}(\boldsymbol{X}) = I\left(\frac{(n-1)S^2}{\sigma_0^2} > \chi^2_\alpha(n-1)\right) \tag{124}$$

is the GLR test of level α.

8. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with an unknown $\theta = (\mu, \sigma^2)^T$ and $n \geq 2$. We test $H_0 : \theta = \theta_0 = (\mu_0, \sigma_0^2)^T$ vs. $H_1 : \theta \neq \theta_0$.
   a) Show that the GLR test statistic is given by:

$$-2\log\Lambda(\boldsymbol{X}) = \frac{n(\bar{X} - \mu_0)^2}{\sigma_0^2} + n\left(\frac{(n-1)S^2}{n\sigma_0^2} - \log\frac{(n-1)S^2}{n\sigma_0^2} - 1\right). \tag{125}$$

Practice Problem VII

b) Show that, under H_0:

$$n\left(\frac{(n-1)S^2}{n\sigma_0^2} - \log\frac{(n-1)S^2}{n\sigma_0^2} - 1\right) \xrightarrow{d} V_1 \sim \chi^2(1). \tag{126}$$

9. Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with an unknown μ and a known σ^2. We test $H_0 : \mu \in [\mu_L, \mu_U]$ vs. $H_1 : \mu \notin [\mu_L, \mu_U]$.

   a) Show that the GLR test of level α is given by:

$$d^{\text{GLR}}(\boldsymbol{X}) = I\left(\bar{X} < \mu_L - c\frac{\sigma}{\sqrt{n}} \quad \text{or} \quad \bar{X} > \mu_U + c\frac{\sigma}{\sqrt{n}}\right), \tag{127}$$

   where the constant $c > 0$ is uniquely determined by the exact equation:

$$\Phi(c) - \Phi\left(-c - \frac{\sqrt{n}(\mu_U - \mu_L)}{\sigma}\right) = 1 - \alpha. \tag{128}$$

Practice Problem VIII

b) Show that the power function of the GLR test is given by:

$$\pi_{d^{\mathbf{GLR}}}(\mu) = 1 - \Phi\left(\frac{\sqrt{n}(\mu_U - \mu)}{\sigma} + c\right) + \Phi\left(\frac{\sqrt{n}(\mu_L - \mu)}{\sigma} - c\right). \tag{129}$$

c) Show that the GLR test is unbiased.

10. For n i.i.d. samples $\boldsymbol{X}_i := (X_{i1}, \ldots, X_{id})^T \sim \mathrm{Multin}(1, \boldsymbol{p})$, suppose we test $H_0 : \boldsymbol{p} = \boldsymbol{p}_0$ vs. $H_1 : \boldsymbol{p} \neq \boldsymbol{p}_0$.

a) Show that the covariance matrix of the MLE $\hat{\boldsymbol{p}}^{\mathrm{MLE}}$ under H_0 is:

$$\mathrm{Var}_{\boldsymbol{p}_0}[\hat{\boldsymbol{p}}^{\mathrm{MLE}}] = n^{-1}\left(\mathrm{diag}(\boldsymbol{p}_0) - \boldsymbol{p}_0\boldsymbol{p}_0^T\right), \tag{130}$$

where $\mathrm{diag}(\boldsymbol{p}_0)$ is a $d \times d$ diagonal matrix with entries $p_{01}, \ldots, p_{0d}$.

Practice Problem IX

b) Let $\boldsymbol{\theta} = (p_1, \ldots, p_{d-1})^T$. Show that the j-th element of the score vector $s_n(\boldsymbol{\theta})$ for $j = 1, \ldots, d-1$ is given by:

$$\left(s_n(\boldsymbol{\theta})\right)_j = n\left(\frac{\hat{p}_j^{\text{MLE}}}{p_j} - \frac{\hat{p}_d^{\text{MLE}}}{p_d}\right). \tag{131}$$

c) Show that the Wald test statistic and Rao's score test statistic (for θ) are both equivalent to Pearson's χ^2 statistic:

$$\sum_{j=1}^{d} \frac{(O_j - E_j)^2}{E_j}, \tag{132}$$

where $O_j = n\hat{p}_j^{\text{MLE}}$ and $E_j = np_{0j}$ represent the observed and expected counts, respectively.

Advanced Practice Problem I

1. Suppose $(X_i, Y_i)^T$ for $i = 1, \ldots, n$ are i.i.d. random vectors from a bivariate normal distribution with parameter $\theta = (\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)^T$. Suppose we test $H_0 : \rho = \rho_0$ vs. $H_1 : \rho \neq \rho_0$ where $-1 < \rho_0 < 1$.
   a) Find the MLEs over Ω_0 and over Ω.
   b) Show that the GLR test statistic is given by:

$$-2 \log \Lambda \left(((X_i, Y_i)^T)_{i=1}^n \right) = n \log \left(\frac{(1 - \rho_0 R)^2}{(1 - \rho_0^2)(1 - R^2)} \right), \tag{133}$$

   where R is the sample correlation coefficient.
   c) Since $\dim(\Omega) = 5$ and $\dim(\Omega_0) = 4$, Wilks' Theorem states that $-2 \log \Lambda \xrightarrow{d} V \sim \chi^2(1)$ under H_0. Show this asymptotic convergence directly by expanding the result in part (b).
2. For a k-dimensional parameter space $\Omega \subseteq \mathbb{R}^k$, consider testing $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$.

Advanced Practice Problem II

a) The Wald test statistic $W(\boldsymbol{X})$ and Rao's score test statistic $S(\boldsymbol{X})$ are defined as:

$$W(\boldsymbol{X}) = (\hat{\theta}_n^{\text{MLE}} - \theta_0)^T \left(\text{Var}_{\theta_0}[\hat{\theta}_n^{\text{MLE}}]\right)^{-1} (\hat{\theta}_n^{\text{MLE}} - \theta_0) \tag{134}$$

and

$$S(\boldsymbol{X}) = s_n(\theta_0)^T \mathcal{I}_n(\theta_0)^{-1} s_n(\theta_0), \tag{135}$$

respectively. Show that, under H_0, both $W(\boldsymbol{X})$ and $S(\boldsymbol{X})$ converge to $\chi^2(k)$ in distribution.

b) Show that the GLR test is consistent; i.e., its power converges to 1 under H_1.

Advanced Practice Problem III

c) Now consider testing $H_0 : \theta = \theta_0$ vs. $H_{1,n} : \theta = \theta_0 + \frac{\delta}{\sqrt{n}}$, where $\delta > 0$ is a known constant. When $Z \sim \mathcal{N}(0,1)$, the distribution of $(Z+\lambda)^2$ is called the *noncentral chi-square distribution* with 1 degree of freedom and noncentrality parameter λ^2, denoted by $\chi^2(1,\lambda^2)$. Show that the asymptotic power of the GLR test is given by:

$$\lim_{n\to\infty} \pi\left(\theta_n := \theta_0 + \frac{\delta}{\sqrt{n}}\right) = P\left(V > \chi^2_\alpha(1)\right), \quad V \sim \chi^2(1, \delta^2 \mathcal{I}_1(\theta_0)). \tag{136}$$

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